

Predicting the seasonality of *Aedes aegypti* in temperate and subtropical regions

Using a stage-structured phenotypically plastic delay-differential equation model



University
of Glasgow

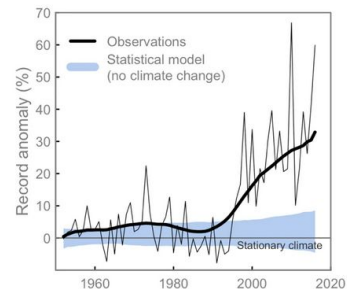
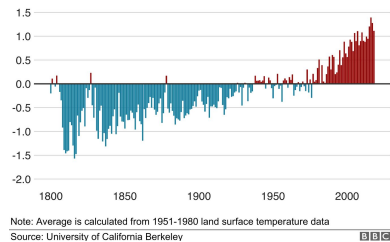
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Christina Cobbold, Steven White, Dominic Brass



UK Centre for
Ecology & Hydrology

Climate change is increasing the range of *Ae. aegypti* further into subtropical climates of the Americas

Temperature and extreme weather events are increasing



Ae. aegypti is spreading in South and North America

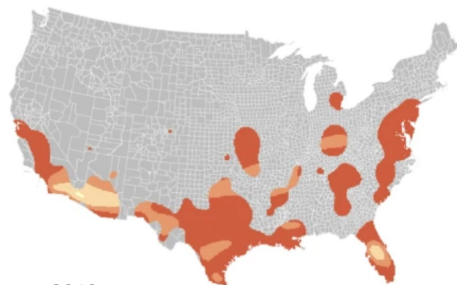
• Mem Inst Oswaldo Cruz. 2020 May 8;115:e190496. doi: [10.1590/0074-02760190496](https://doi.org/10.1590/0074-02760190496)

***Aedes aegypti* spreading in South America: new coldest and southernmost records**

[Alejandra Rubio](#)^{1,2,*}, [María Victoria Cardo](#)^{1,2}, [Darío Vezzani](#)^{1,3}, [Anibal Eduardo Carbajo](#)^{1,2}

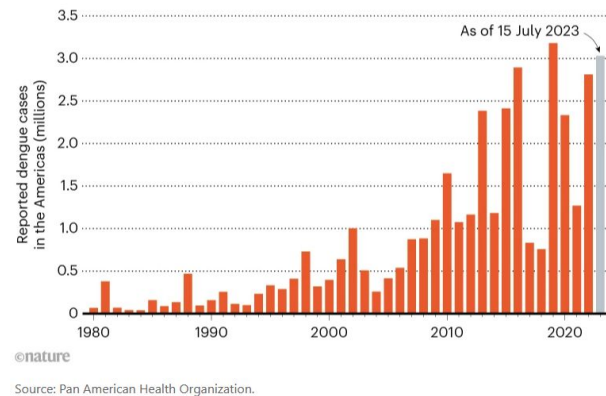
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PMCID: PMC7207151 PMID: [32401999](https://pubmed.ncbi.nlm.nih.gov/32401999/)



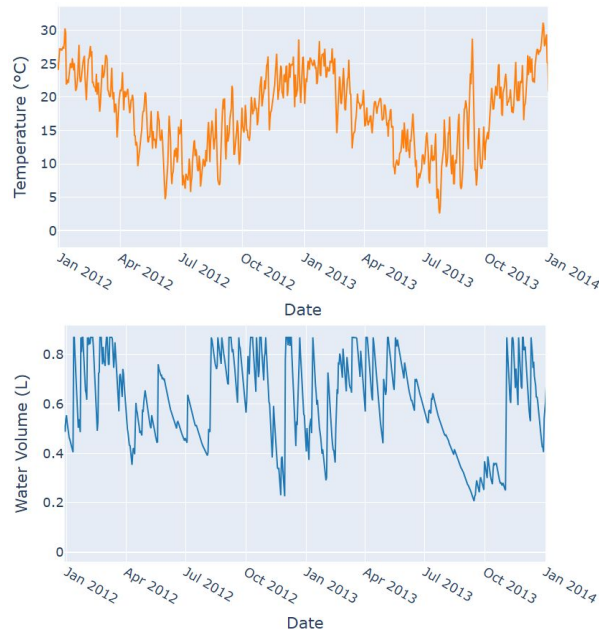
Kraemer 2019

Dengue cases are increasing as the virus is spreading to naive populations

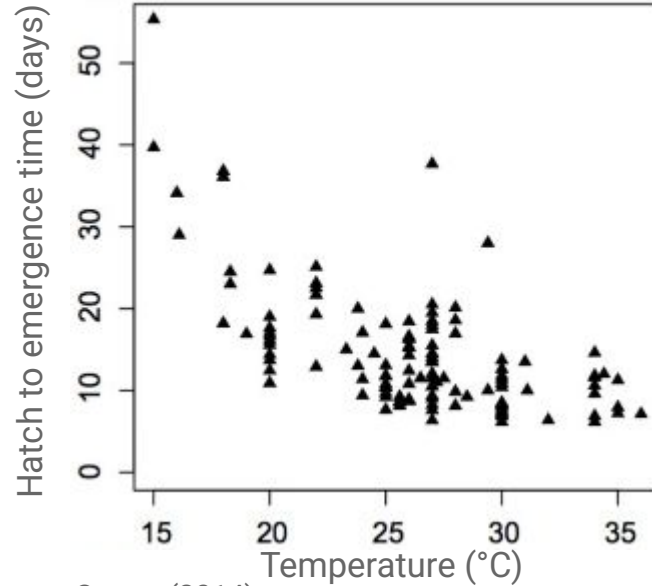


To effectively model dengue fever at these range edges the delays from hatch to biting are important

Ae. aegypti is invading **more variable climates**

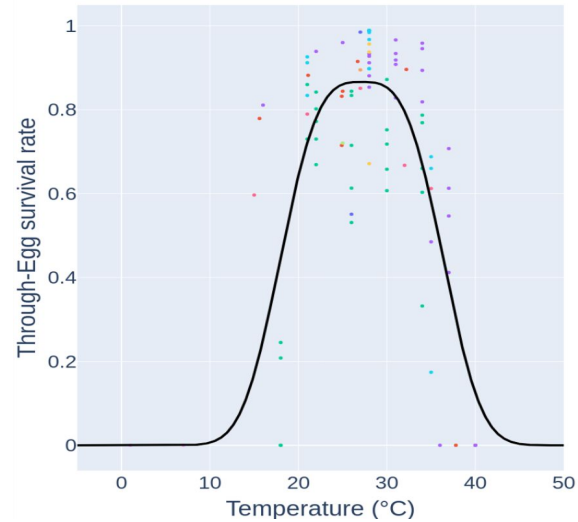


The **delays** are environmentally dependent



Couret (2014)

Traits, such as survival, **change** during development



Córdoba, Argentina

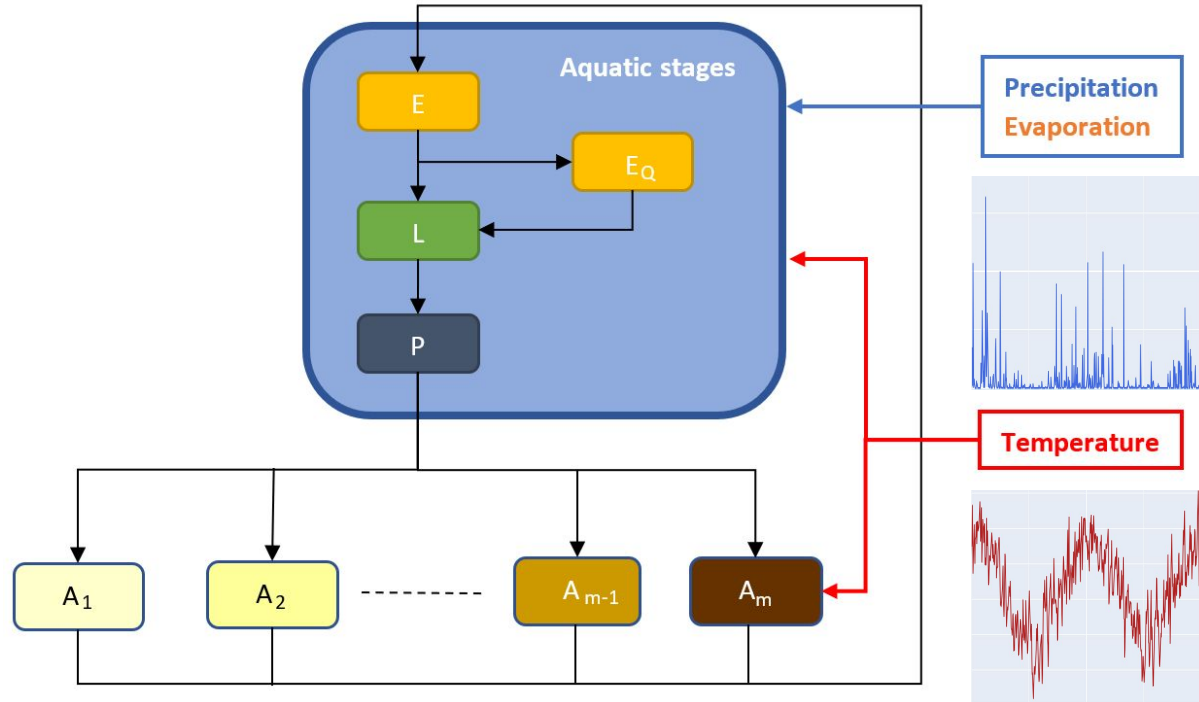
Building a stage-structured time delayed phenotypically plastic model for *Ae. aegypti*

Role of vector phenotypic plasticity in disease transmission as illustrated by the spread of dengue virus by *Aedes albopictus*

Dominic P. Brass , Christina A. Cobbold, Bethan V. Purse, David A. Ewing, Amanda Callaghan & Steven M. White

Nature Communications 15, Article number: 7823 (2024) | [Cite this article](#)

We have 5 stages: developing eggs, quiescent eggs, larvae, pupae, and adults (which are divided into size-classes)



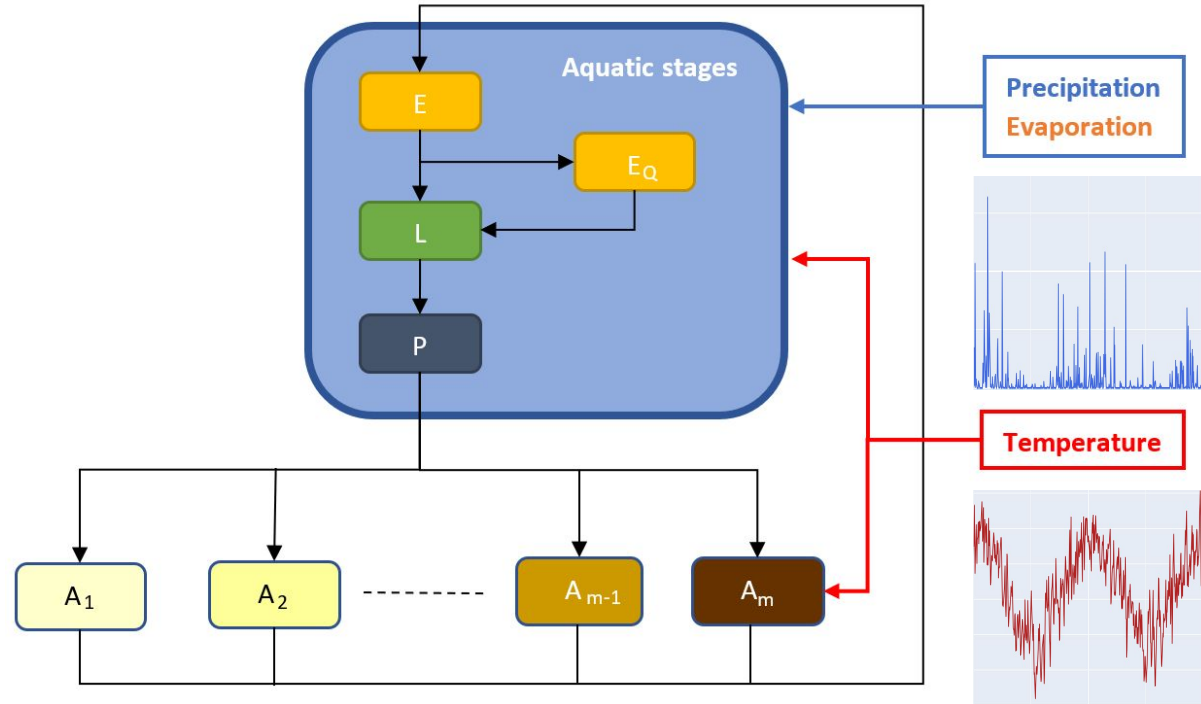
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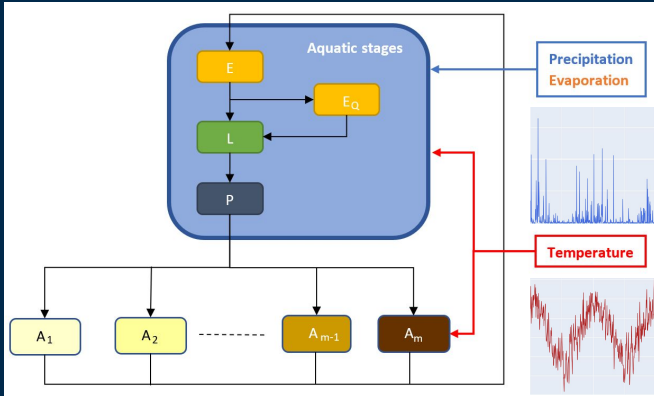


Note: as this model focuses on subtropical and temperate regions so only aims to model naturally filled containers

Building a stage-structured time delayed phenotypically plastic model for *Ae. aegypti*

The equations for these stages are given by delay-differential equations:

$$\begin{aligned}
 \frac{dE}{dt} &= \overbrace{R_E(t)}^{\text{Recruitment}} - \overbrace{M_E(t)}^{\text{Maturation}} - \overbrace{\delta_E(t)E(t)}^{\text{Mortality}}, \\
 \frac{dE_Q}{dt} &= R_{E_Q}(t) - M_{E_Q}(t) - \delta_E(t)E_Q(t), \\
 \frac{dL}{dt} &= R_L(t) - M_L(t) - \delta_L(t)L(t), \\
 \frac{dP}{dt} &= R_P(t) - M_P(t) - \delta_P(t)P(t), \\
 \frac{dA_i}{dt} &= R_{A_i}(t) - \delta_A(t)A(t).
 \end{aligned}$$



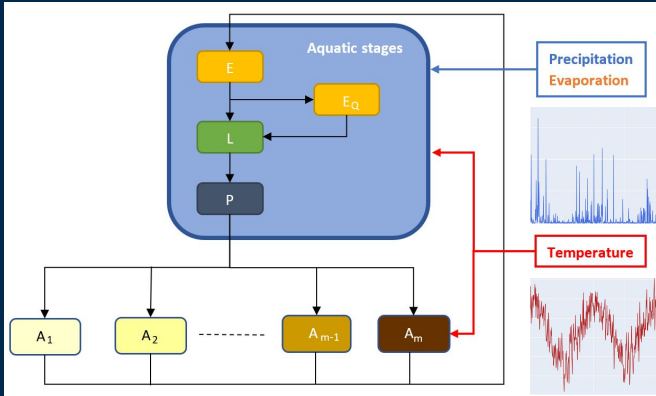
Building a stage-structured time delayed phenotypically plastic model for *Ae. aegypti*

These maturation terms induce lags in the system:

$$M_E(t) = R_E(t - \tau_E(t)) \frac{g_E(t)}{g_E(t - \tau_E(t))} S_E(t)$$

The stage lengths produce delays in the system

- The delays are variable due to the environmental conditions
- Both from temperatures across the stages and food availability in the larval stage



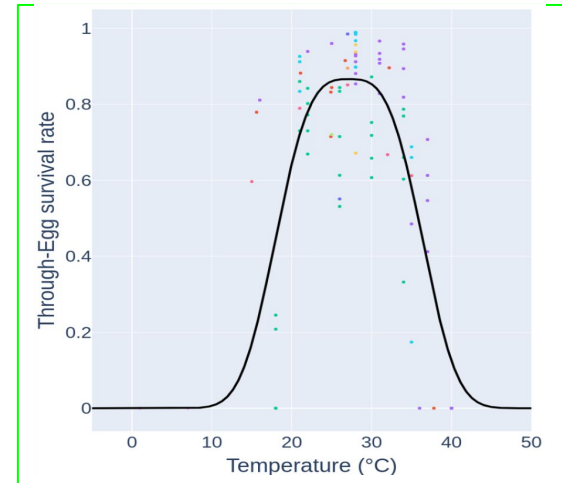
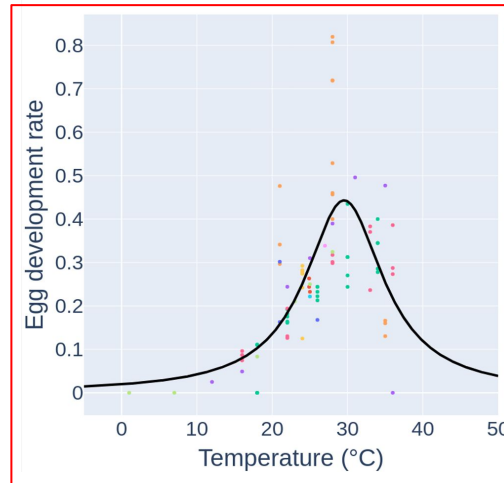
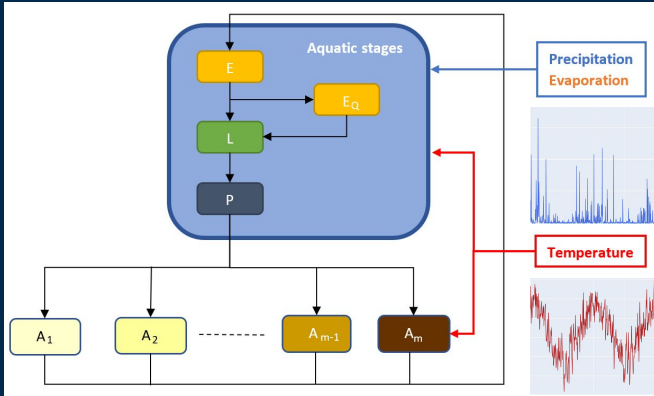
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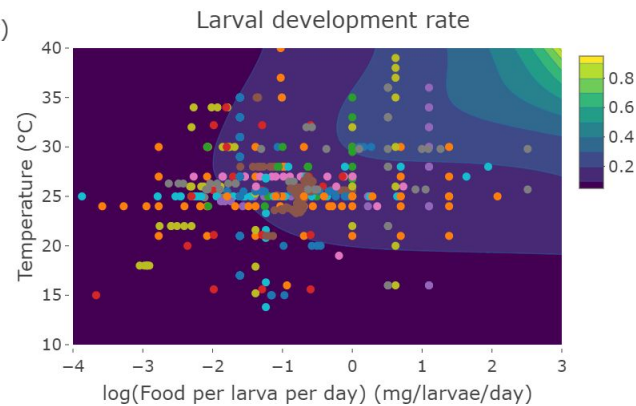
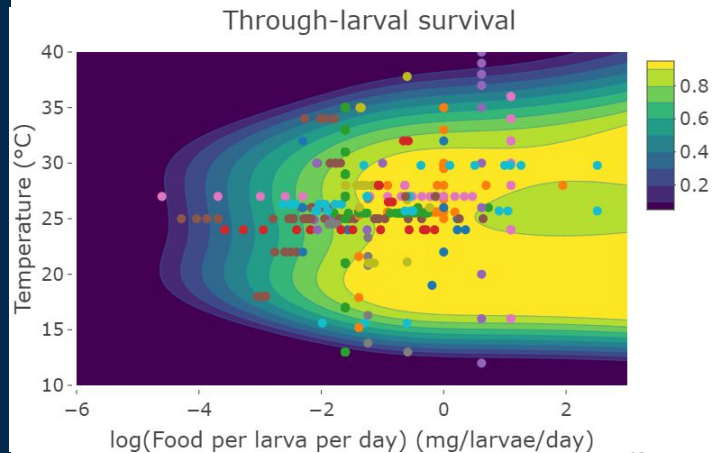
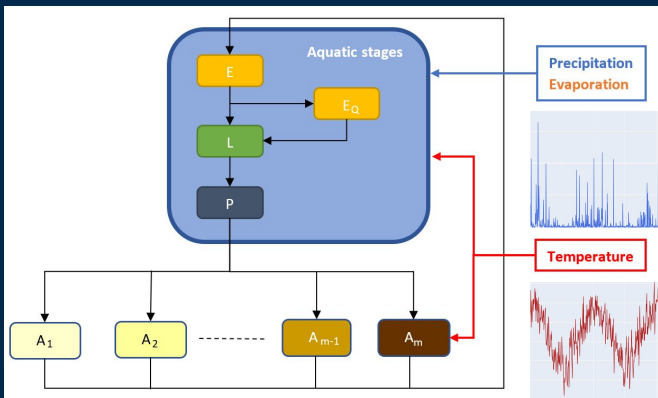
We parameterise reaction norms (environment- trait relationships) for life traits across each stage:

- A large number of constant temperature lab experiments are used to parametrise these reaction norms



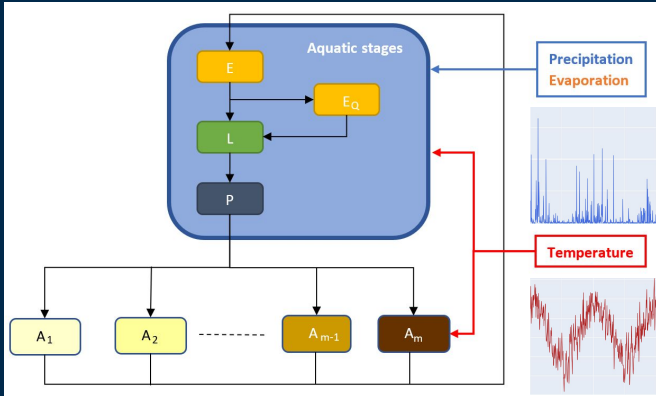
Building a DDE model for *Ae. aegypti* -Larval environment

Larvae compete for food in the larval stage, which acts as a density dependent population regulation mechanism. The larval food is used to create reaction norms for development rate and through-larval survival rate

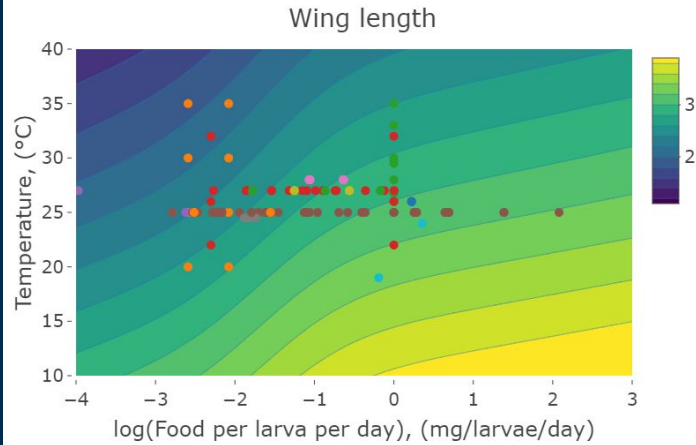


Building a DDE model for *Ae. aegypti*

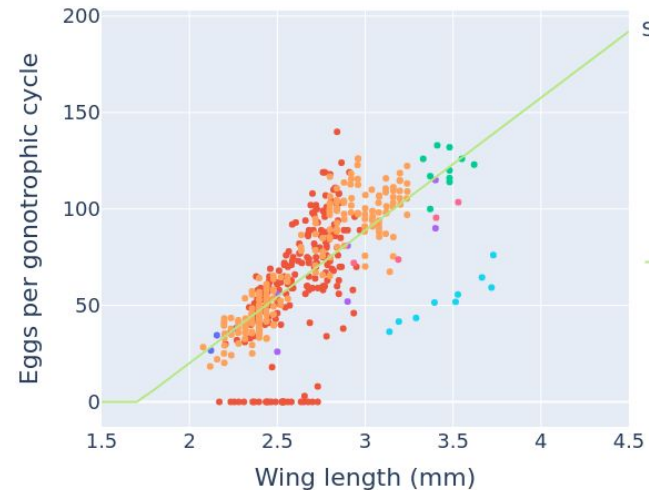
-Phenotypic plasticity



The larval environment creates a delayed response in adult wing length



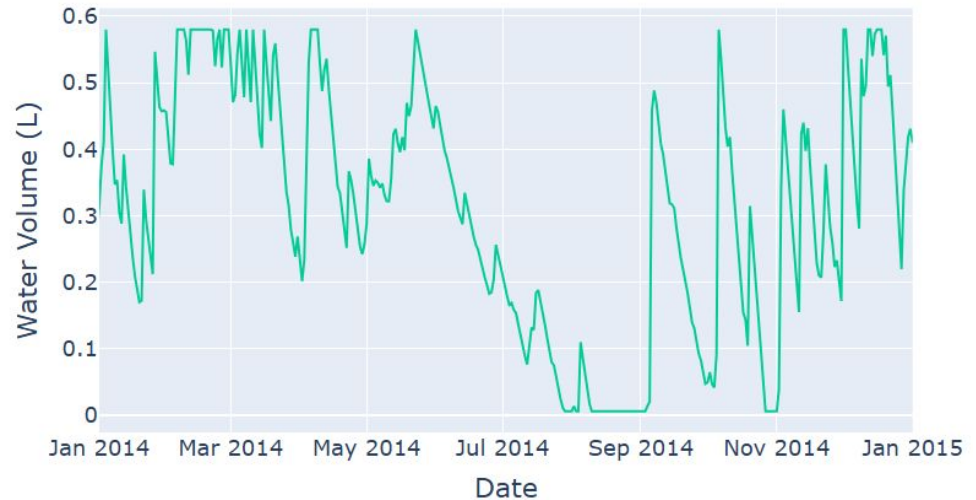
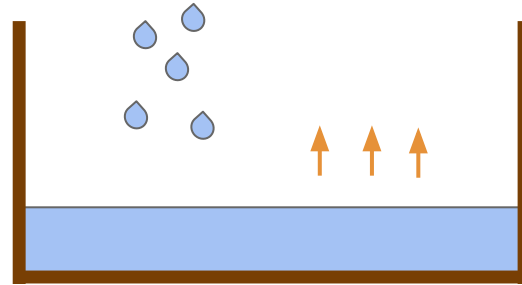
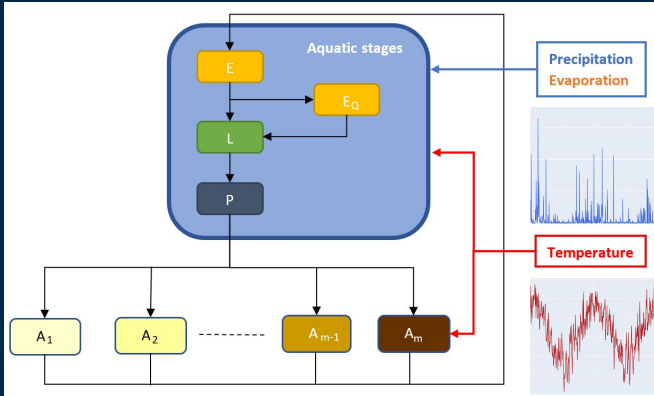
Wing length in adults can produce a **2-fold increase in eggs per gonotrophic cycle**



Building a DDE model for *Ae. aegypti* -Container

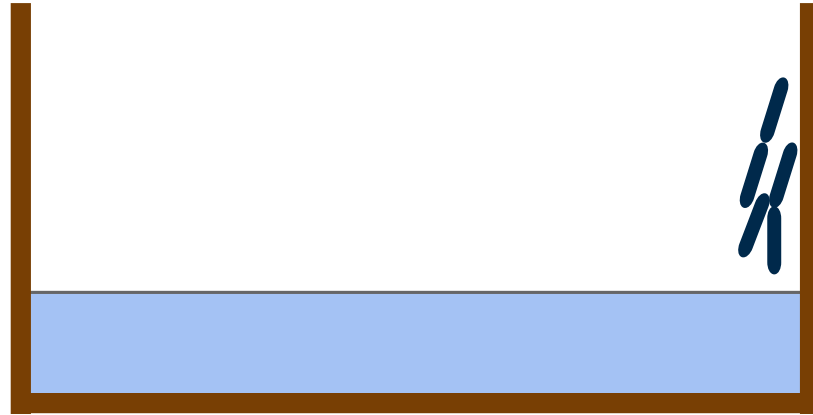
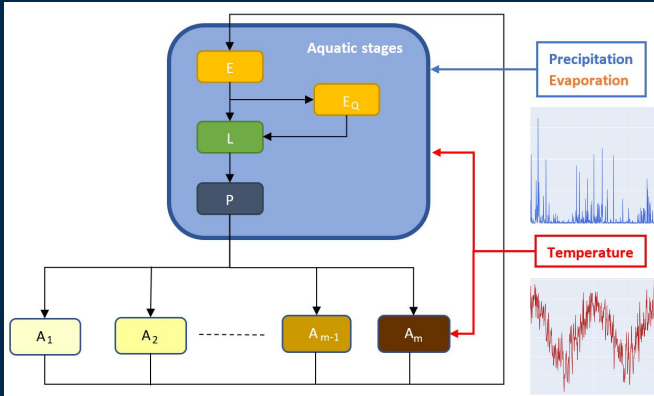
For the simulations we model one container selected to be the size of a **flower pot** (from larval surveys and egg laying experiments)

The **water volume** changes due to **precipitation** and **evaporation**



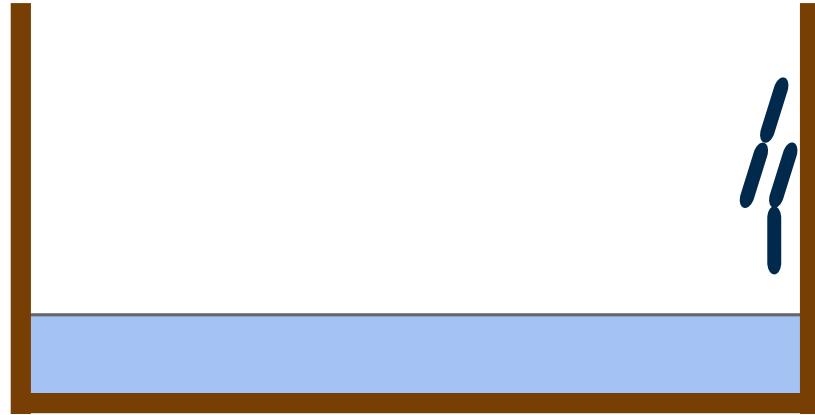
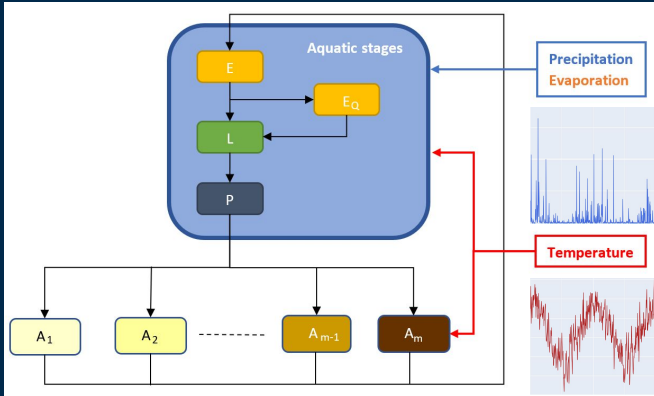
Building a DDE model for *Ae. aegypti* -Hatching

- Eggs get **laid near the water level**
- Half of submerged eggs hatch upon development
- All non-hatching eggs go into the quiescent eggs
- Half of submerged (quiescent) eggs hatch per day



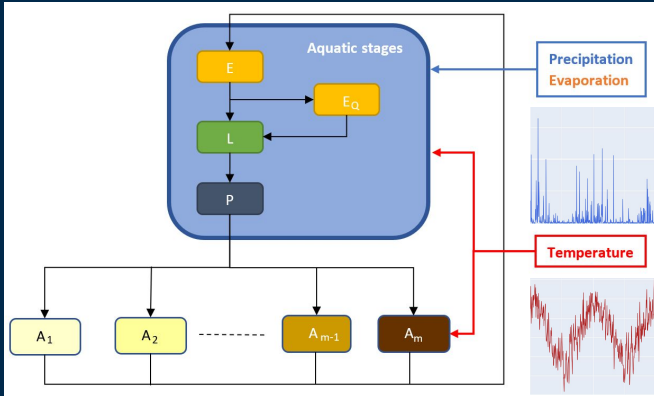
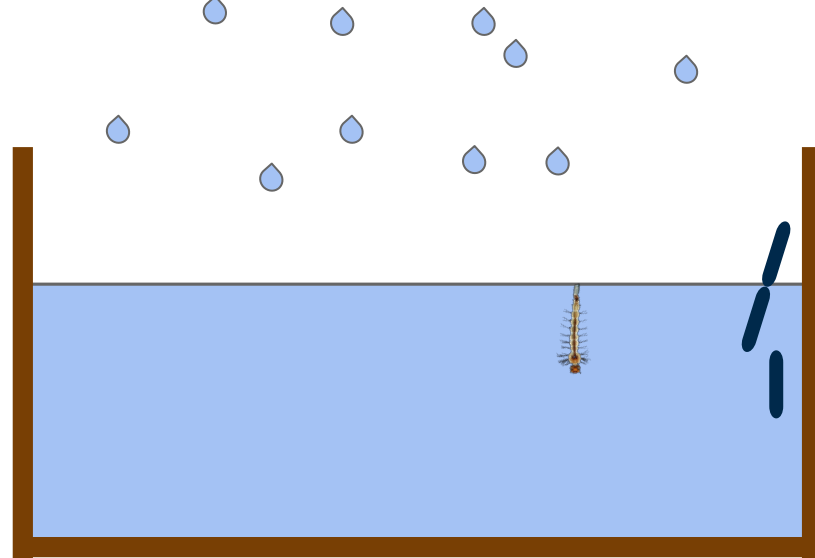
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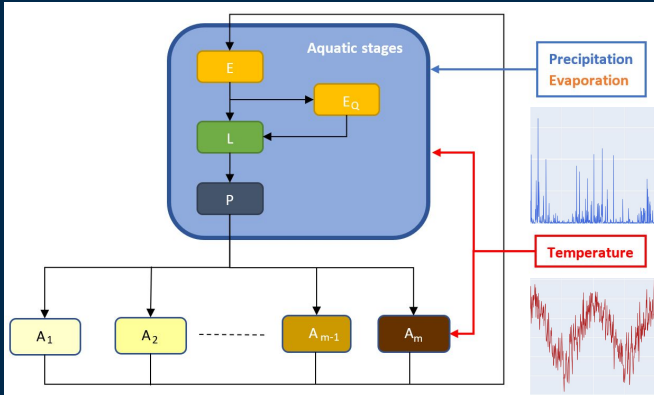
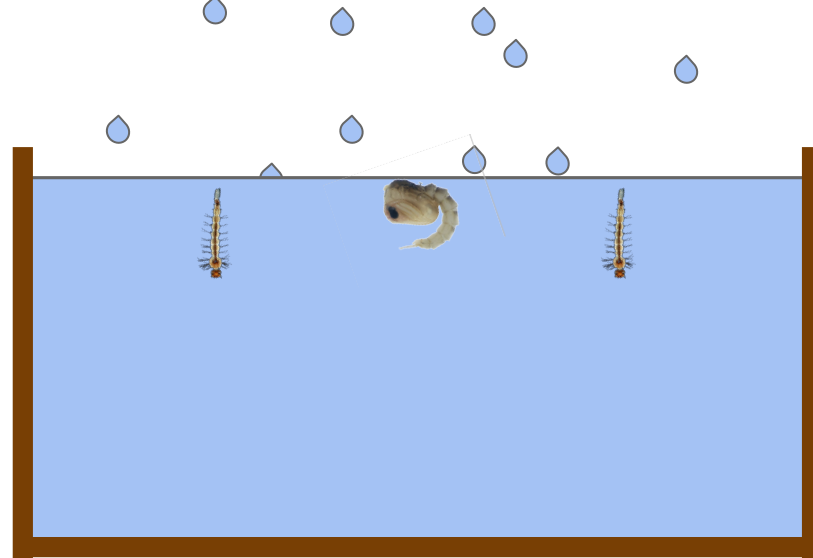
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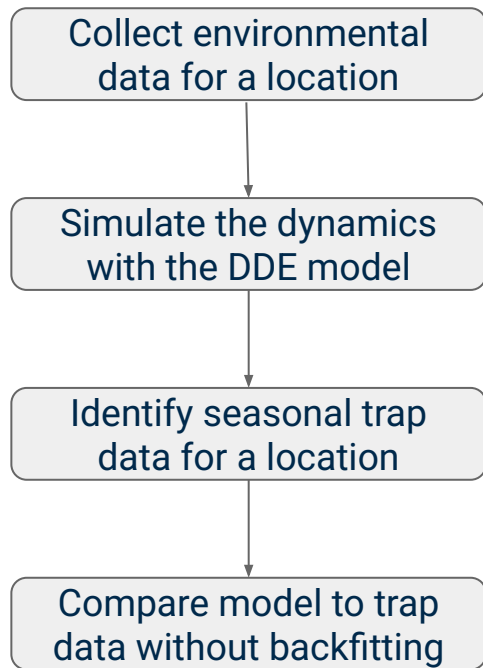


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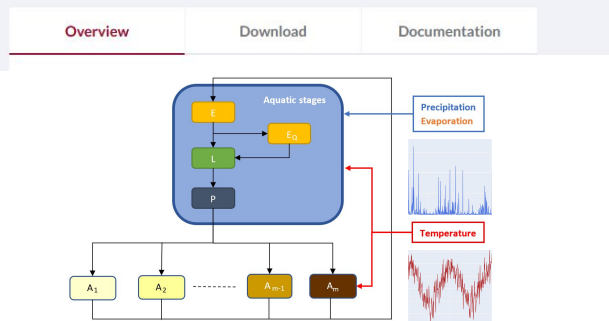
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Running validation data



ERA5-Land hourly data from 1950 to present

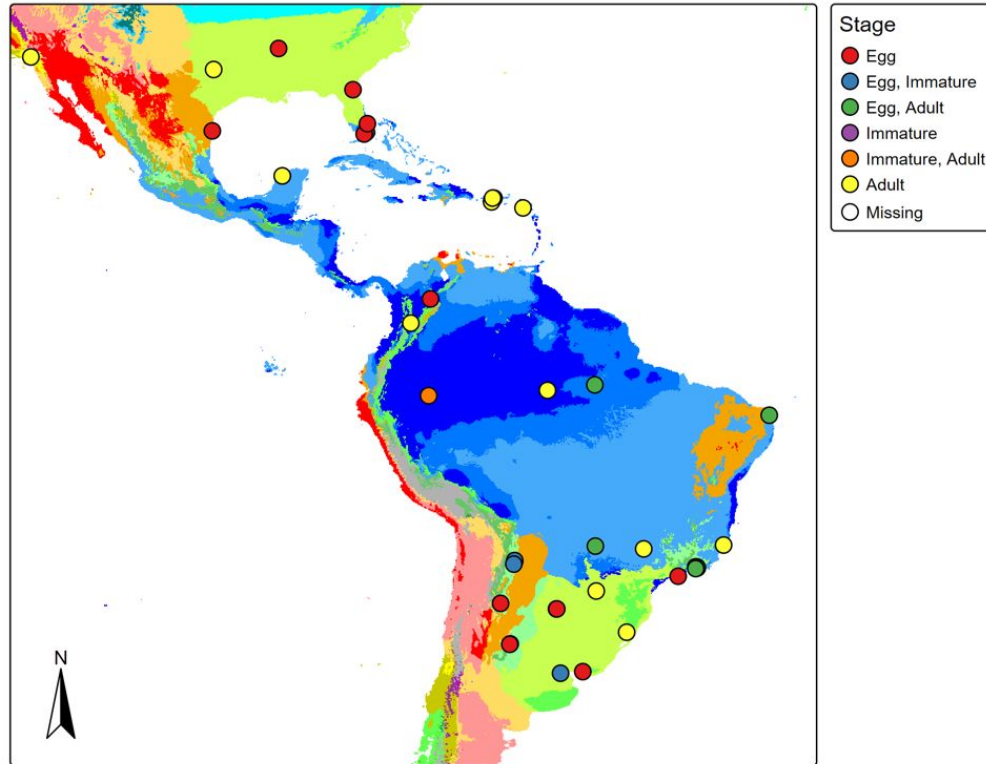


Longitudinal trap data from adult traps or ovitraps

Average daily temperature, cumulative daily precipitation and evaporation



We have validated the model across the Americas (without back-fitting)

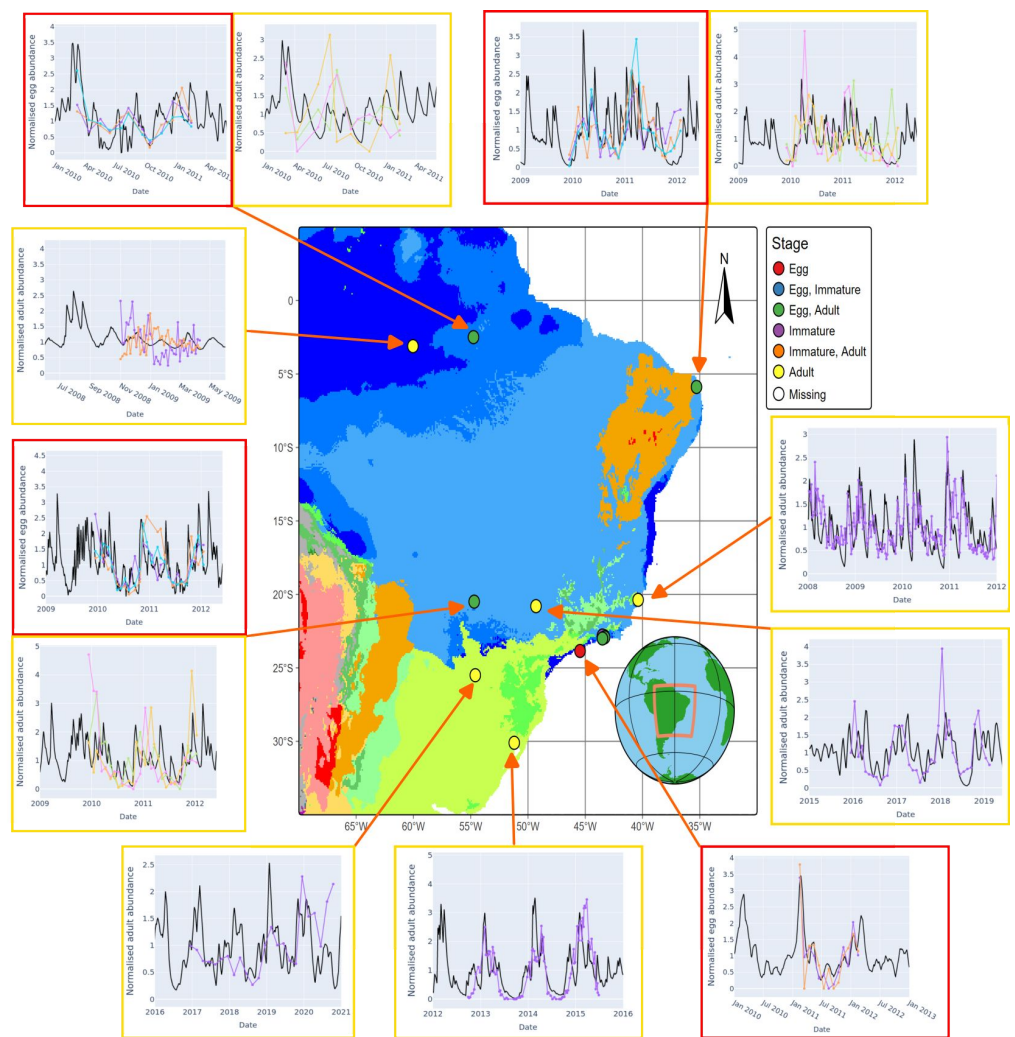


- We have **41 trap data sets**
- These span the **Köppen climate classification**:
 - Af - Tropical Rainforest (4)
 - Am - Tropical Monsoon (7)
 - As/Aw - Tropical Savanna (12)
 - **Cwa/Cfa - Humid subtropical (16)**
 - Cfb - Oceanic (1)
 - Bsh - Semi-arid (1)

The model works well
in across many
climates (without
back-fitting)

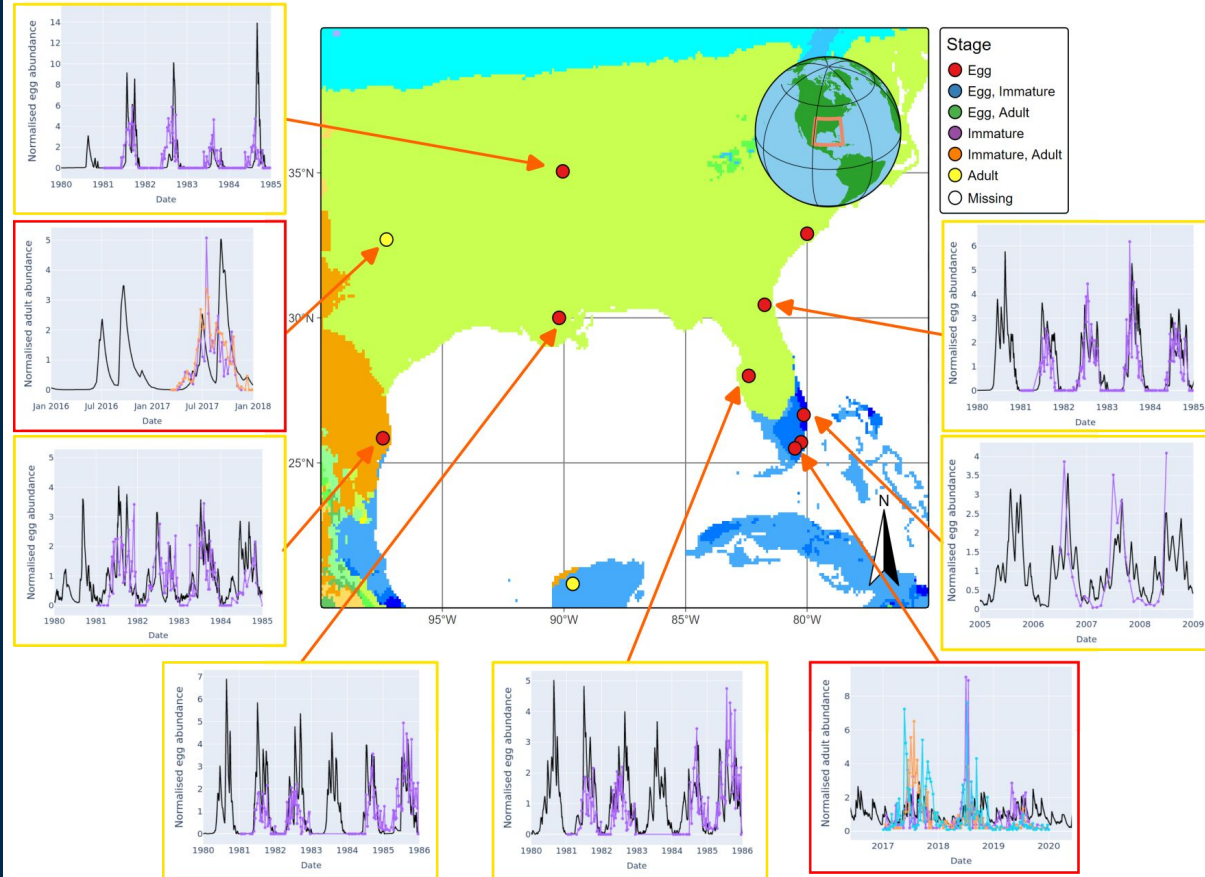
Although the model is designed to
model the range edges, the model
works well across many climates

Model simulations in black
against trap data in other colours



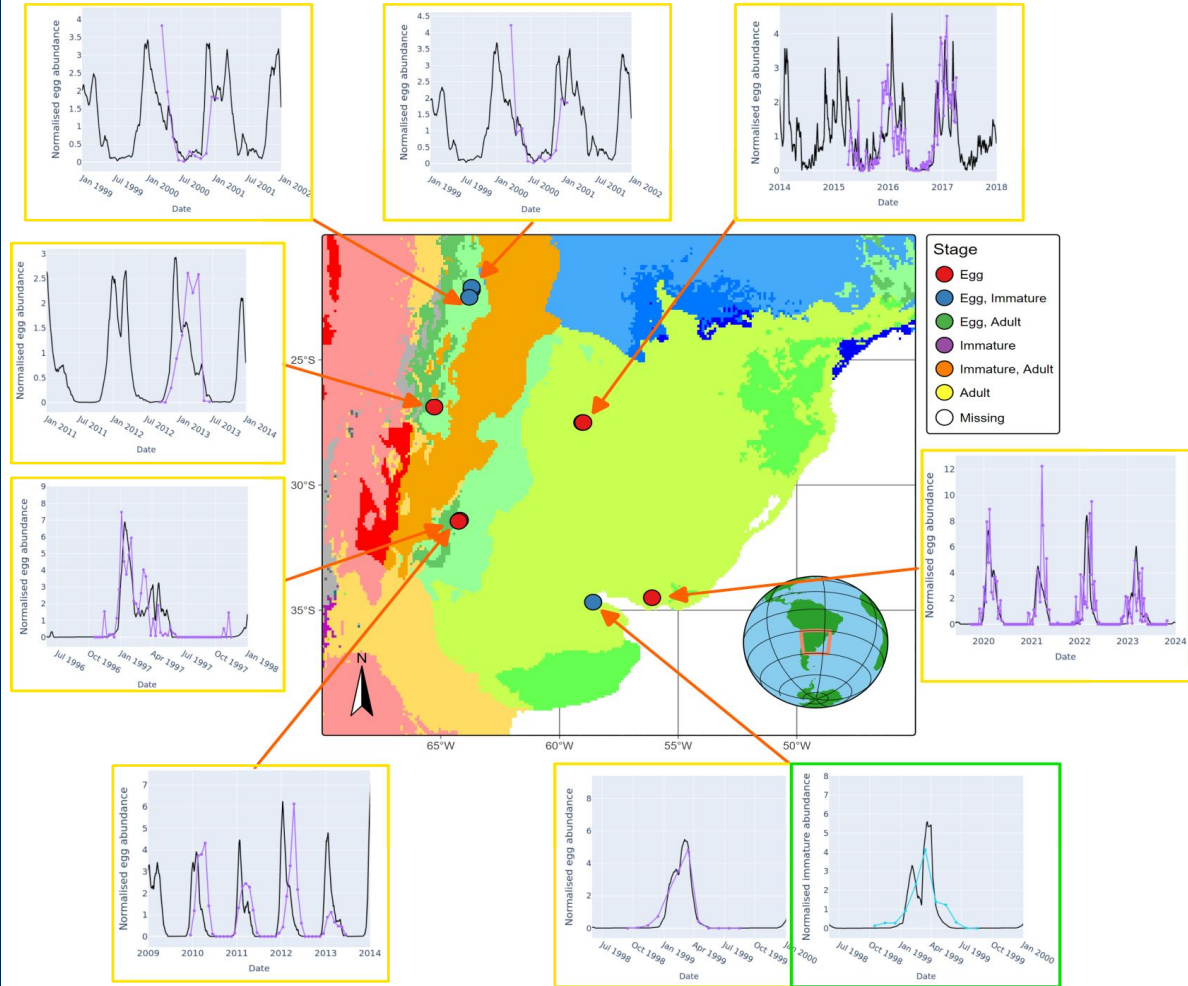
We have validated the model across temperate and subtropical climates in the Americas (without back-fitting)

Model simulations in black
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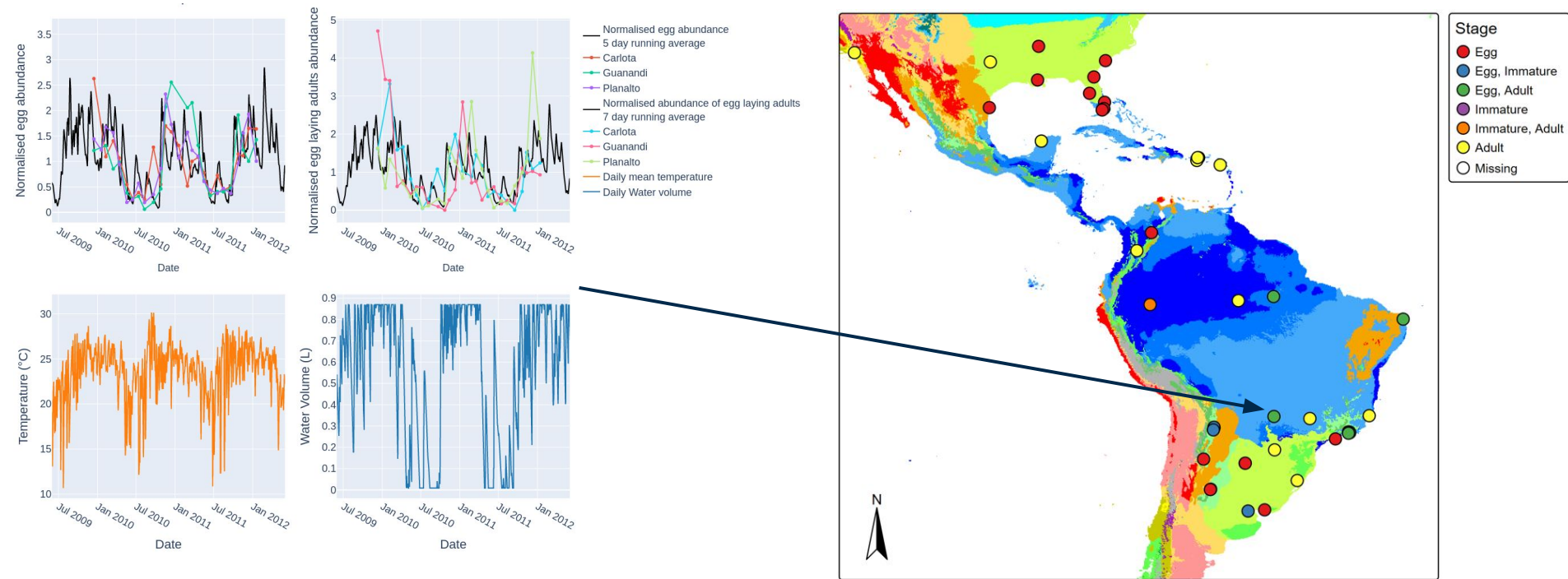


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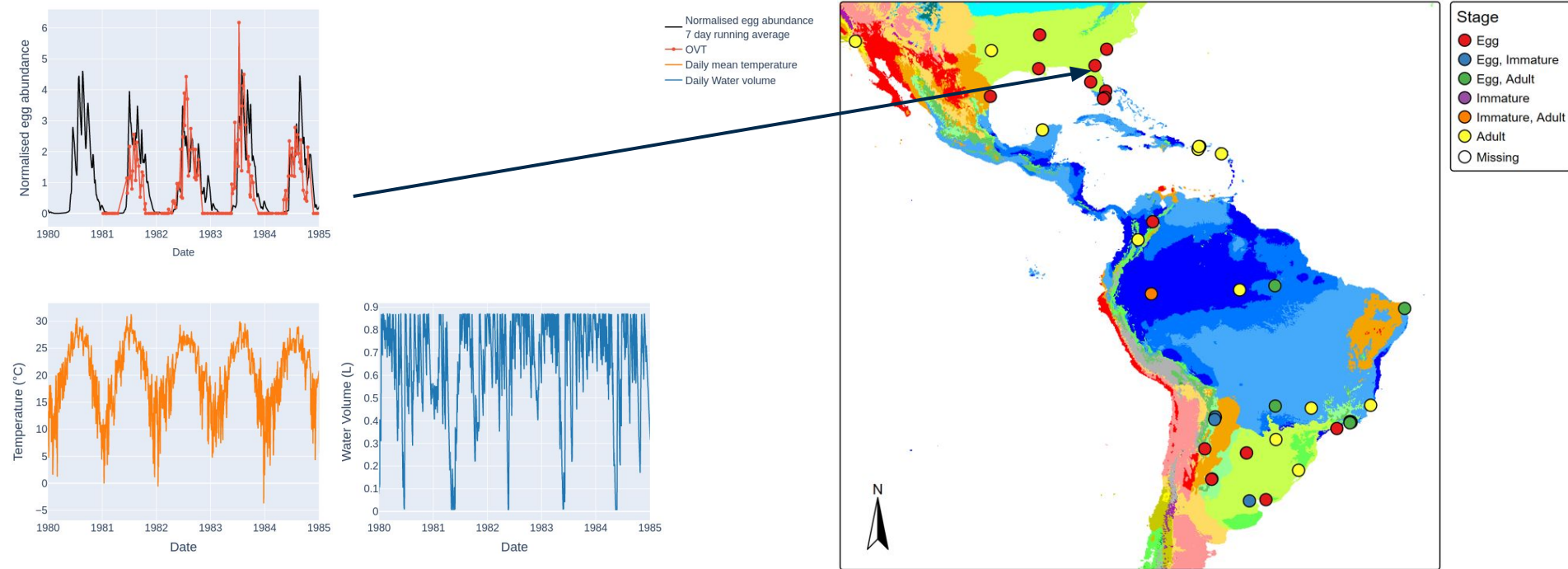


Works well in large portions of their current range: tropical savanna climate



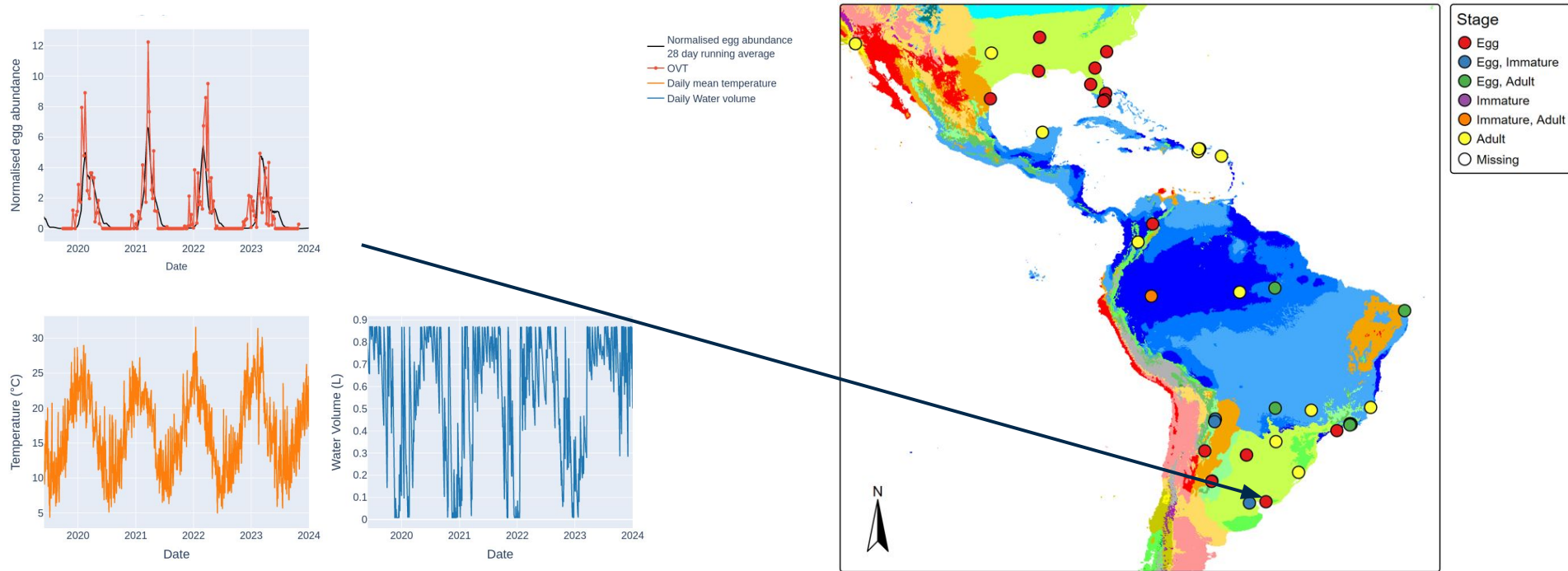
Campo Grande, Mato Grosso do Sul, Brazil: Egg and Adult trap data 2010-2011, Codeco 2015

Works well in range edge: humid subtropical climates



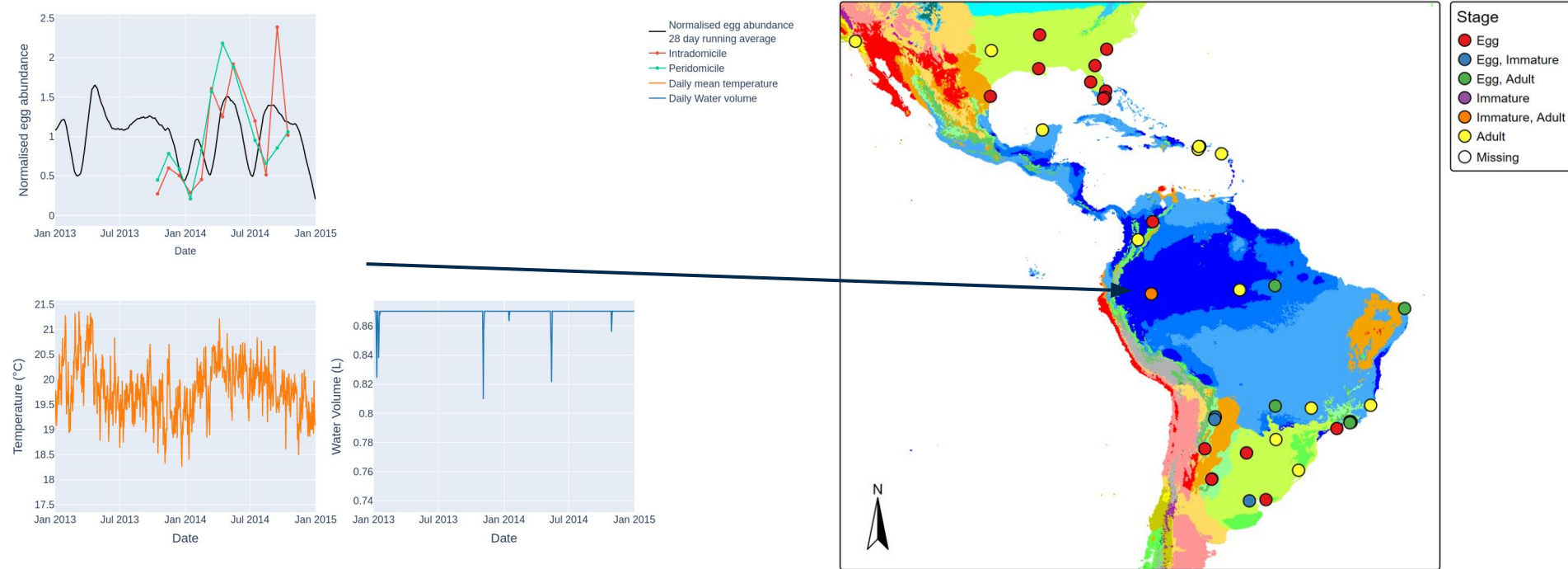
Jacksonville, Florida, United States of America: Egg trap data 1981-1984, Focks 1993

Works well in range edge: humid subtropical climates



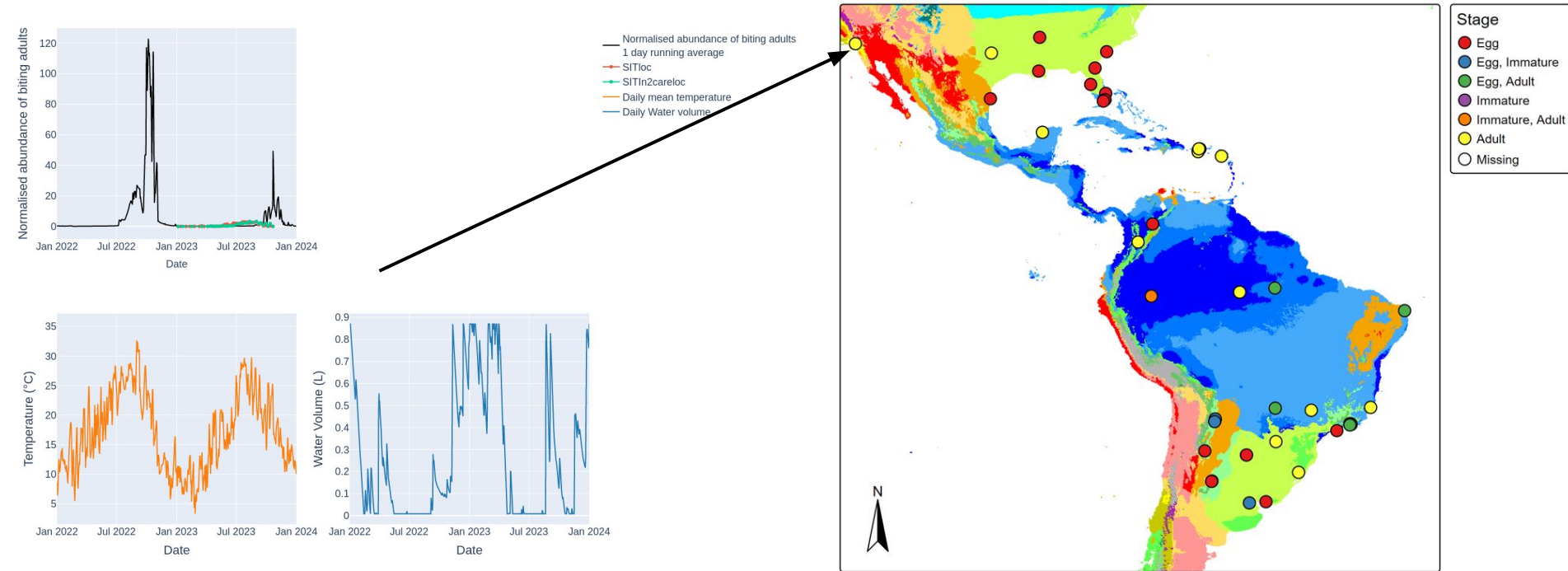
San Antonio, Canelones, Uruguay: Egg trap data, 2020-2023, Vidal 2024

Areas the model is not designed for: tropical rainforest/tropical monsoon climates



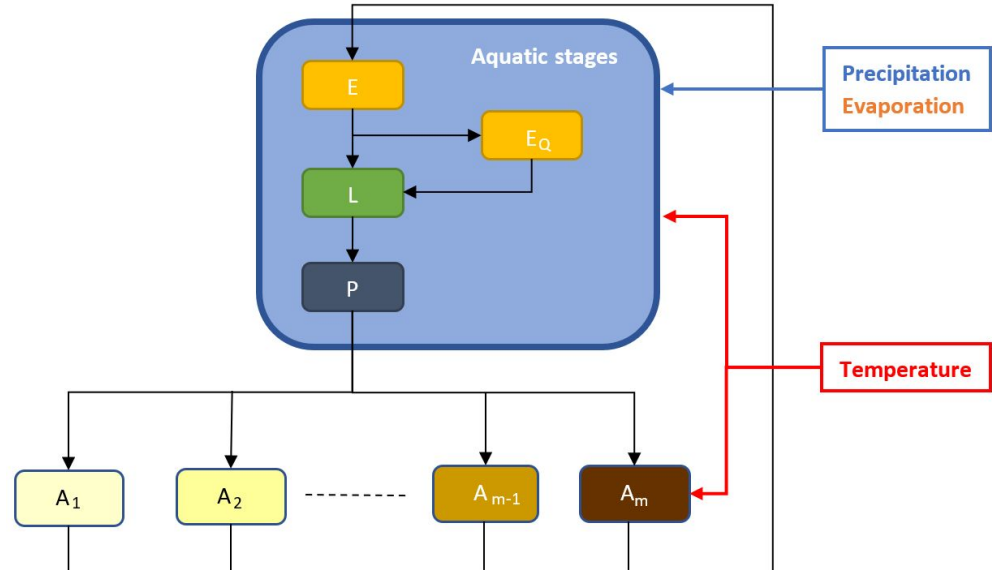
Bucaramanga, Santander, Columbia, Egg trap data 2014, Ruiz 2018

Areas the model is not designed for: semi-arid/arid climates



Conclusion/Main takeaways

- ***Ae. aegypti* is expanding its range** into sub-tropical regions of North and South America
- **Delay differential equations** with delayed phenotypic plasticity in the adult size can **accurately predict *Ae. aegypti* seasonality**
- In the **range edge** these results can be used to predict the seasonality of the population **across life-stages**

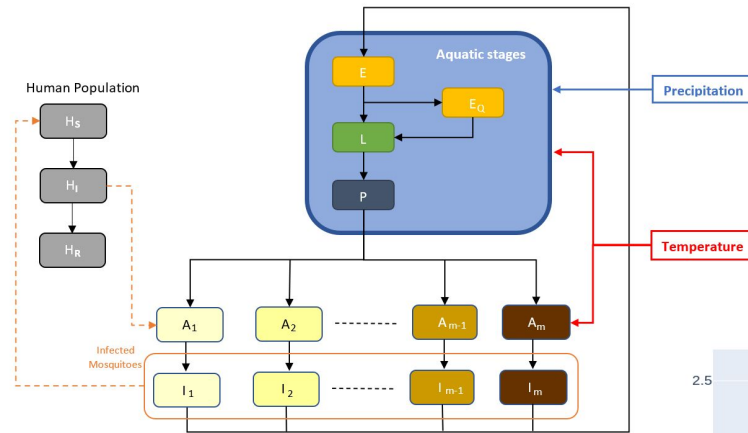


Current and future work on dengue fever outbreaks

Expanding the model of *Ae. aegypti* to a dengue SEIR model.

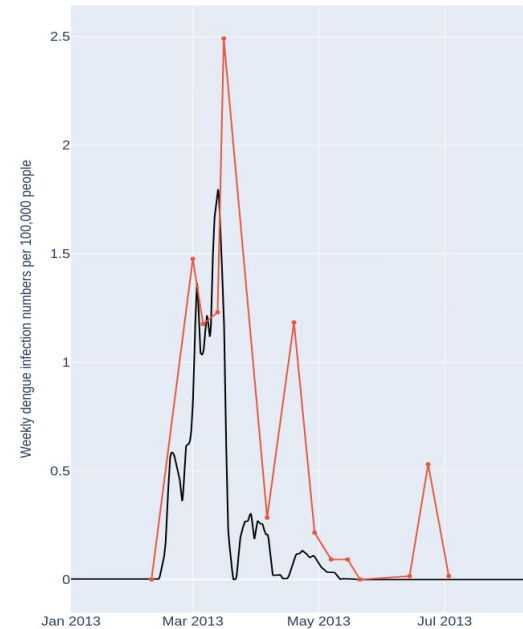
Time delays occur through the Extrinsic Incubation Period (EIP)

Can we predict the shape and severity of dengue fever outbreaks in the range limits?



Preliminary model results in black against case data in orange for dengue infections in Córdoba, temperate Argentina

Dengue Cases



Any questions?

Feel free to contact me:

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Special thanks to Christina Cobbold, Steven White, and Dominic Brass for support and EPSRC for the funding of my PhD



Engineering and
Physical Sciences
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